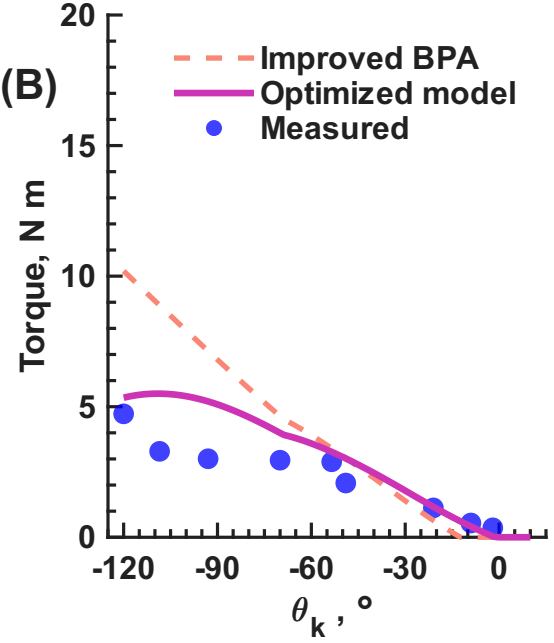
(A)

(B)


Improved BPA
Optimized model
Measured
Torque, N m
20
15
10
5
0
-120
-90
-60
-30
0
θ_k, °